

Post-Quantum Cryptography

Security for the Quantum Era

Presentation Outline

01

Cryptography Basics

02

RSA Security

03

Quantum Threats

04

PQC Overview

05

PQC Families

06

NIST Standards

07

PQC Analysis

08

Current Limits

09

Future Outlook

10

Conclusion

Fundamentals of Cryptography

Cryptography is the cornerstone of digital security, safeguarding information from unauthorized access and tampering.



Where Cryptography Matters



Banking

Transactions & accounts



Messaging

Private conversations



Internet

HTTPS & online activity



Digital Signatures

Document & software verification

Symmetric vs. Asymmetric Cryptography

Feature	Symmetric	Asymmetric
Key	Same key	Public/private pair
Speed	Faster	Slower
Examples	AES	RSA ECC

The Quantum Realm: Bridging Theory to Threat

Quantum computing leverages principles of quantum mechanics to perform computations far beyond the capabilities of classical machines, posing both immense opportunities and significant challenges for existing cryptographic systems.

Core Quantum Concepts

These fundamental principles enable quantum computers to process information in unique ways.



Qubits

Unlike classical bits, qubits can represent 0 and 1 simultaneously.



Superposition

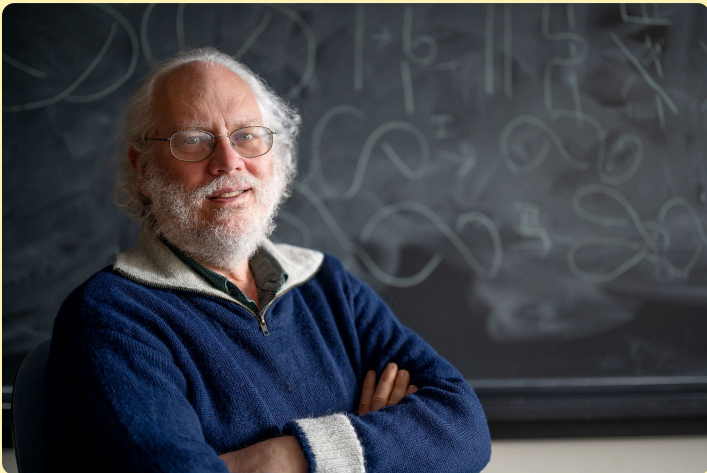
Allows qubits to exist in multiple states at once, enabling parallel computation.



Entanglement

Links qubits together so the state of one affects the others instantly.

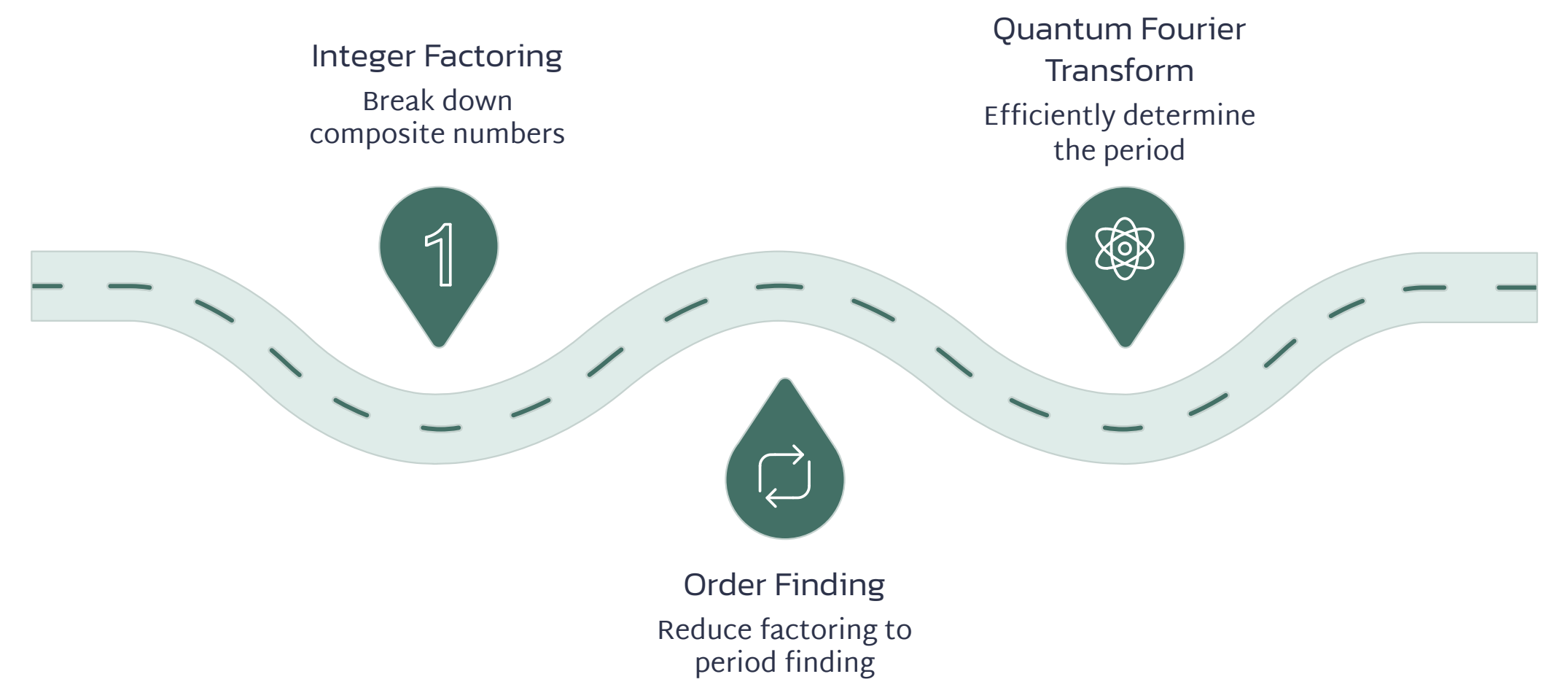




Shor's Algorithm: A Quantum Threat

Developed by Peter Shor in 1994, this algorithm can efficiently factor integers and solve discrete logarithms, threatening to break widely used cryptographic systems like RSA and Diffie-Hellman.

Powerful quantum computers could compromise classical public-key cryptography.



If the period r is found, it can reveal the prime factors of n and compromise systems like RSA and Diffie-Hellman.

The Quantum Threat Emerges

Classical public-key cryptography relies on mathematical problems that are computationally difficult for traditional computers. Quantum computing changes this landscape dramatically.



Complexity Comparison

Quantum algorithms drastically reduce the time needed to solve problems previously considered intractable.

Algorithm	Complexity
General Number Field Sieve (GNFS)	Sub-exponential
Shor’s Algorithm	Polynomial

 **Insight:** Polynomial-time scaling makes quantum attacks dramatically more practical for large cryptographic keys.



Why This Matters: RSA & ECC at Risk

Quantum computers threaten RSA and ECC because they can solve factoring and discrete logarithm problems far more efficiently.

Introduction to Post-Quantum Cryptography (PQC)

Post-Quantum Cryptography (PQC) represents a new generation of cryptographic algorithms designed to withstand attacks from future quantum computers, safeguarding our digital world from emerging threats.

Unlike classical cryptography, PQC leverages different mathematical problems believed to be intractable even for quantum machines.

Feature	RSA/ECC (Classical)	PQC (Quantum-Safe)
Quantum Safe	No	Yes
Key Size	Small	Larger
Speed	Mature (generally faster)	Varies (can be slower)
Standardization	Established	Emerging (NIST processes underway)

The transition to PQC is crucial to maintain long-term digital security and data privacy.

Introduction to Post-Quantum Cryptography (PQC)

PQC is designed to resist quantum attacks while still running on classical hardware.



Classical Compatibility

Built to work on today's hardware.



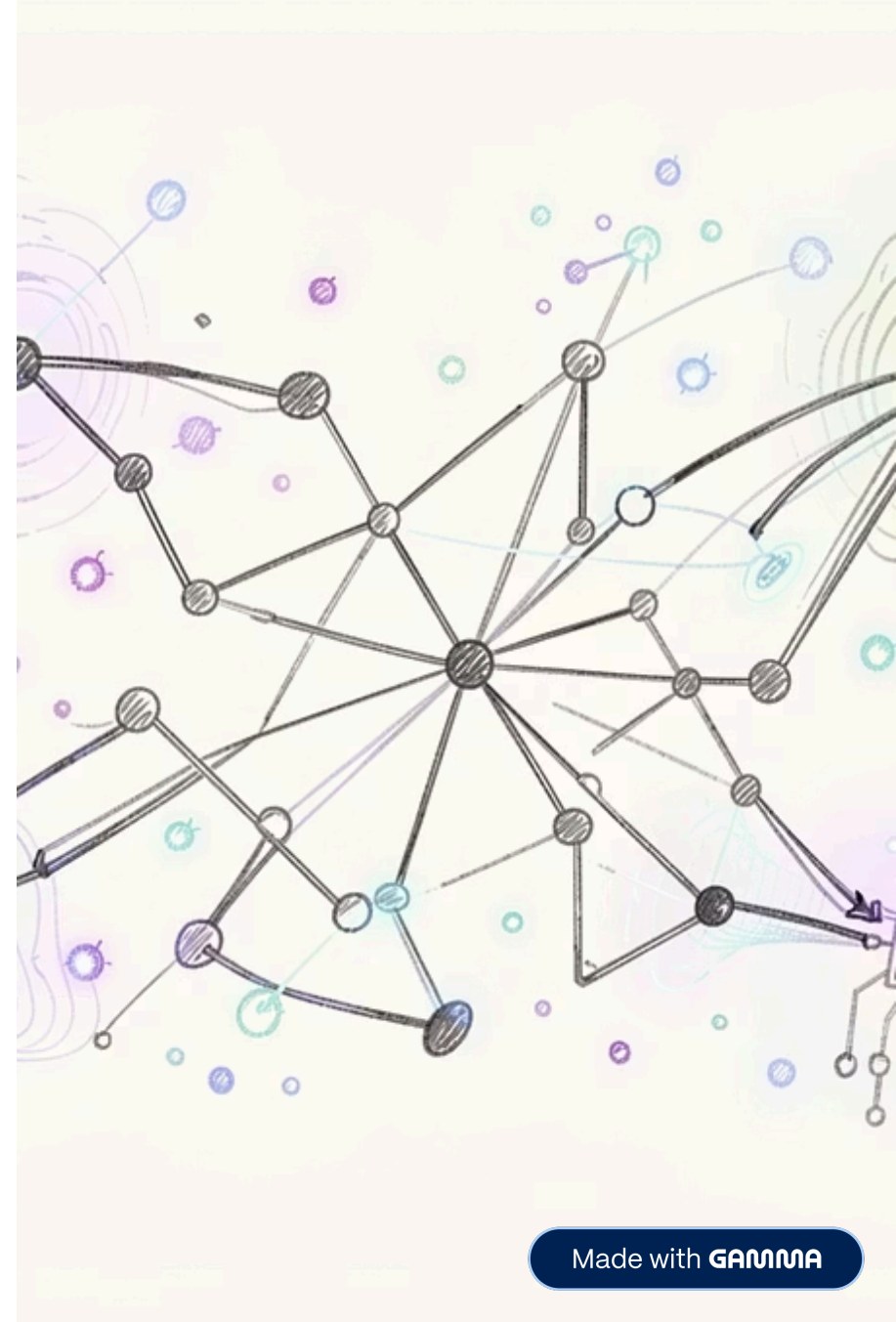
Successor to RSA/ECC

Created to replace quantum-vulnerable public-key cryptography.



New Hardness Assumptions

Relies on problems quantum computers can't efficiently solve.



Five Main Families of Post-Quantum Cryptography

PQC uses different mathematical problems to stay secure against quantum attacks.

Family	Hard Problem	Examples
Lattice-based	SVP, LWE	Kyber, Dilithium
Code-based	Syndrome decoding	McEliece
Hash-based	Hash collision resistance	SPHINCS+, XMSS
Multivariate	Multivariate equations	Rainbow, GeMSS
Isogeny-based	Elliptic-curve isogenies	SIDH, SIQsign

These families form the foundation of quantum-safe cryptography.

Lattice-Based Cryptography

Lattice-based cryptography is the leading family of Post-Quantum Cryptography (PQC) and a major focus of NIST standardization.

$$\Lambda(\mathbf{B}) = \left\{ \sum_{i=1}^n z_i \mathbf{b}_i \mid z_i \in \mathbb{Z} \right\}$$

SVP

Find the shortest non-zero lattice vector.

LWE

Recover a secret from noisy linear equations.

MLWE

An efficient algebraic variant of LWE.

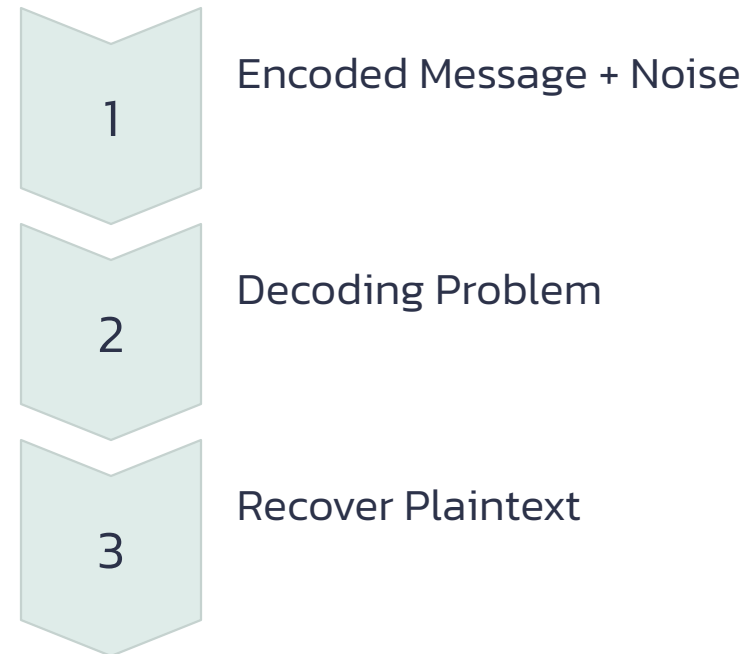
KYBER

DILITHIUM

Key examples for **Key Encapsulation** and **Digital Signatures**.

Code-Based Cryptography

Code-based cryptography relies on error-correcting codes and the hardness of the Syndrome Decoding Problem.



Advantages

- Strong theoretical foundation
- Quantum-resistant

Disadvantages

- Very large key sizes
- Can be less efficient

McEliece is the most well-known and long-standing secure code-based cryptosystem.

Hash-Based Cryptography

A conservative post-quantum approach built on cryptographic hash functions.



Hash Foundations

Security relies on collision-resistant hash functions.



Conservative Security

One of the most studied PQC families.



Merkle Trees

Enable efficient verification with a single public key.



SPHINCS+

NIST-standardized stateless hash-based signatures.



Merkle trees support efficient signature verification.

Multivariate & Isogeny-Based Cryptography

Multivariate Public-Key Cryptography

Based on solving multivariate polynomial equations over finite fields, centered on the **MQ problem**.

Rainbow, a NIST finalist, was broken by practical attacks and later deprecated.

Isogeny-Based Cryptography

Uses elliptic curve isogenies, with **SIKE** known for small keys and elegant structure.

A 2022 classical attack broke SIKE, removing it from NIST standardization. **SQIsign** remains under research.

 **Elegant mathematics $\neq$ guaranteed security.** SIKE is a reminder that beautiful constructions can still fail under scrutiny.

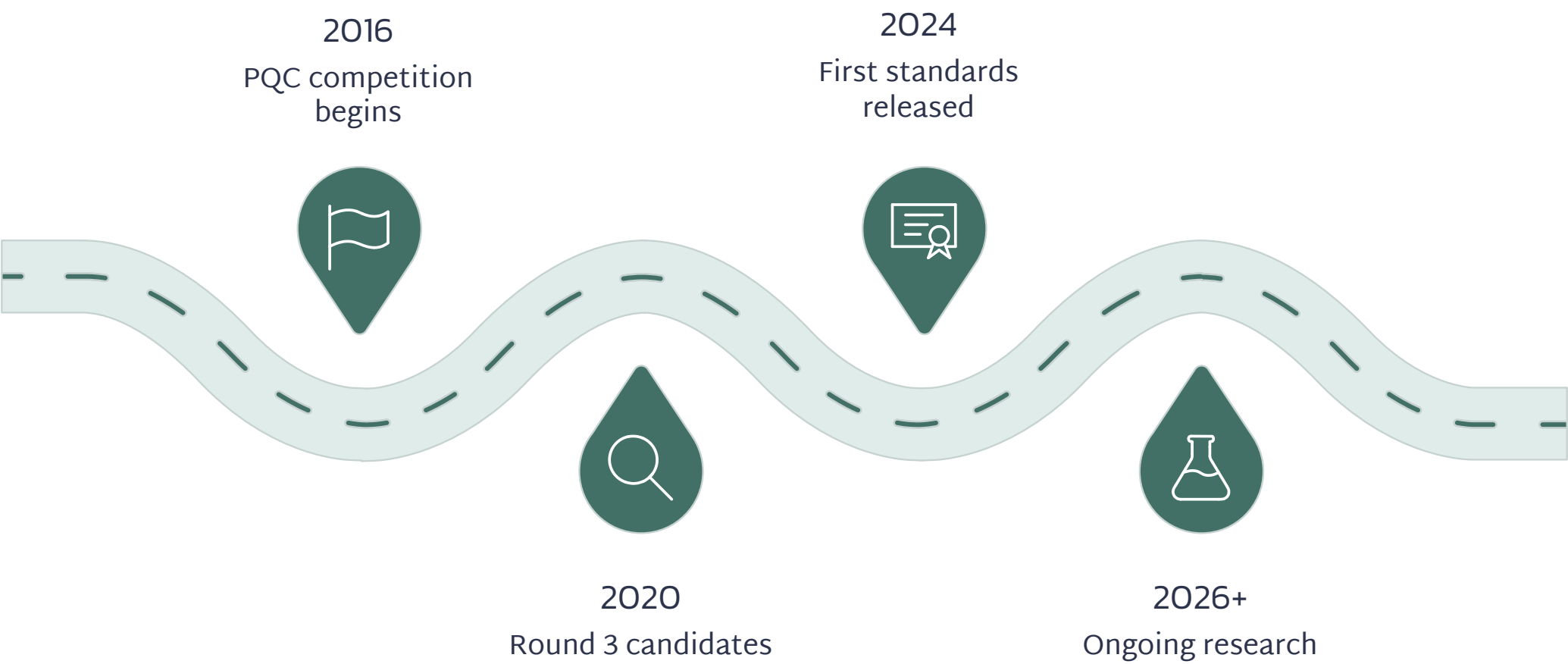
NIST Standardization: Securing the Future

NIST began its Post-Quantum Cryptography (PQC) standardization effort in **2016** to select quantum-resistant algorithms through rigorous evaluation.

Current PQC Standards

Standard	Algorithm	Description
FIPS 203	ML-KEM	Key establishment
FIPS 204	ML-DSA	Digital signatures
FIPS 205	SLH-DSA	Hash-based signatures
FIPS 206	Falcon	Compact lattice signatures

PQC Standardization Timeline



NIST’s PQC program has moved from competition to standards, with more research still underway.

Why Lattice Schemes Dominate

Lattice-based cryptography leads NIST PQC because it combines strong security with practical performance for real-world use.

Speed

Fast execution for high-volume cryptographic operations.

TLS Ready

Small keys and strong efficiency make integration practical.

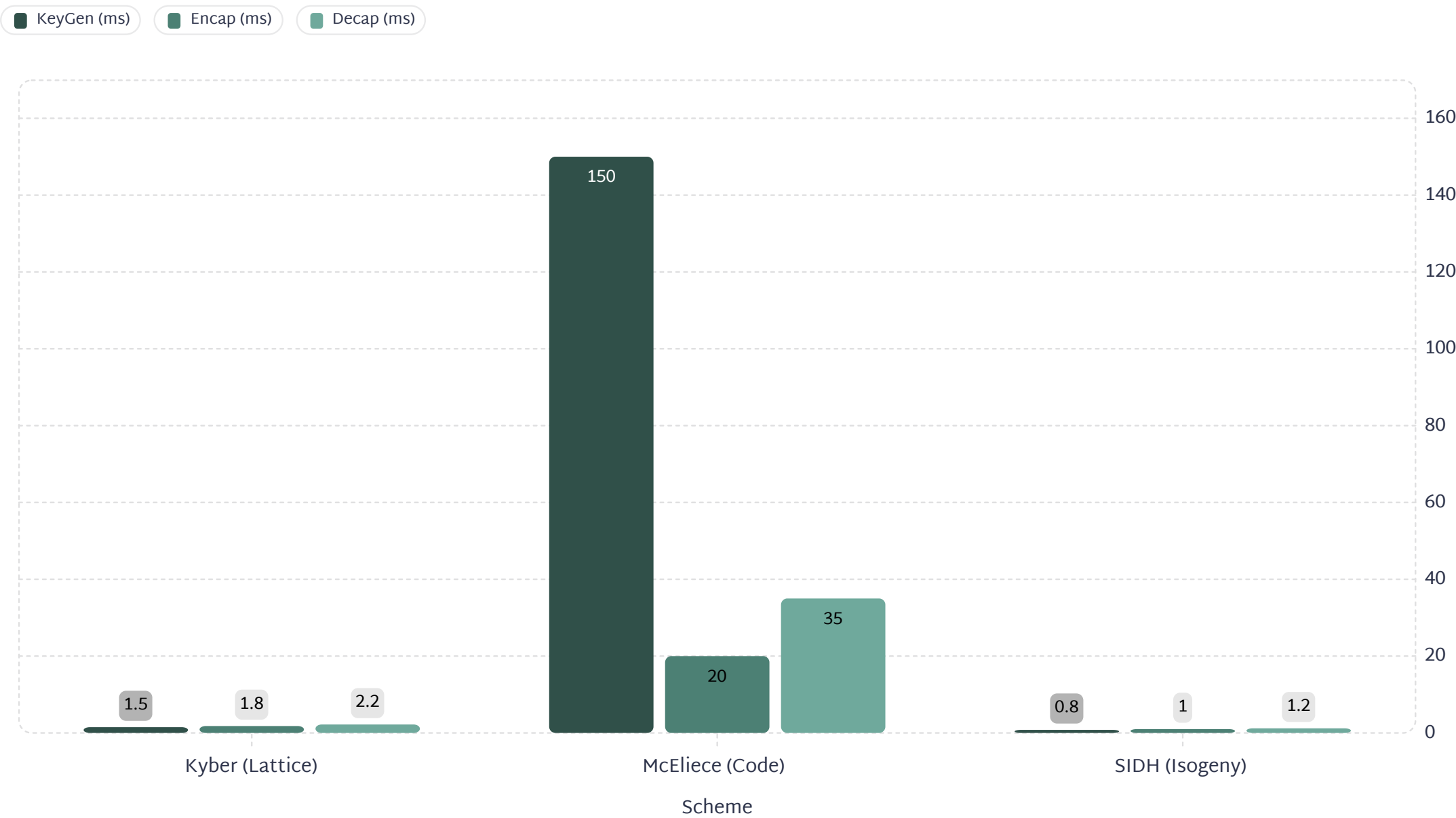
NTT Acceleration

Number Theoretic Transform boosts polynomial arithmetic.

MLWE Foundation

Built on a well-studied problem with quantum-resistant security.

KEM Performance Comparison



Kyber is far more practical than McEliece, while SIDH is no longer viable.

Analyzing PQC Algorithms

Evaluating Post-Quantum Cryptography (PQC) means checking whether an algorithm is secure, efficient, and practical against classical and quantum attacks.



Security Reductions

Formal proof that security rests on a hard mathematical problem.



Parameter Selection

Chooses key sizes and security levels for the right balance of strength and efficiency.



Performance

Measures speed, memory use, and bandwidth.



Attack Models

Tests resistance to classical, quantum, and side-channel attacks.



Mathematical Tools

Uses algebra and number theory to assess hardness and attack surface.

SECURITY REDUCTIONS: The Foundation of PQC

Security reductions show that breaking a PQC scheme is as hard as solving a known hard problem.

PQC Schemes and Their Hard Problems

Scheme	Underlying Hard Problem
ML-KEM	Module Learning With Errors (MLWE)
McEliece	Syndrome Decoding Problem
SPHINCS+	Collision Resistance of Hash Functions

QROM extends these proofs to quantum adversaries.

Parameter Selection: Fine-Tuning for Quantum Security

Parameter selection is a key step in PQC deployment. It balances **security**, **performance**, and **memory use** against classical and quantum attacks.

Key parameters include:

- **Polynomial Degree (n):** Larger values usually increase security and cost.
- **Modulus (q):** Affects arithmetic and the hardness of LWE-style problems.
- **Module Rank (k):** Tunes security-efficiency tradeoffs in module-lattice schemes.

ML-KEM-768 is tuned for NIST Security Level 3, targeting at least **192 bits classical** and **96 bits quantum** security.

NIST PQC Security Levels

Level	Classical	Quantum
1	≥ 128	≥ 64
3	≥ 192	≥ 96
5	≥ 256	≥ 128

These levels provide a clear target for quantum-resistant parameter choices.

Performance Comparison: Key and Signature Sizes

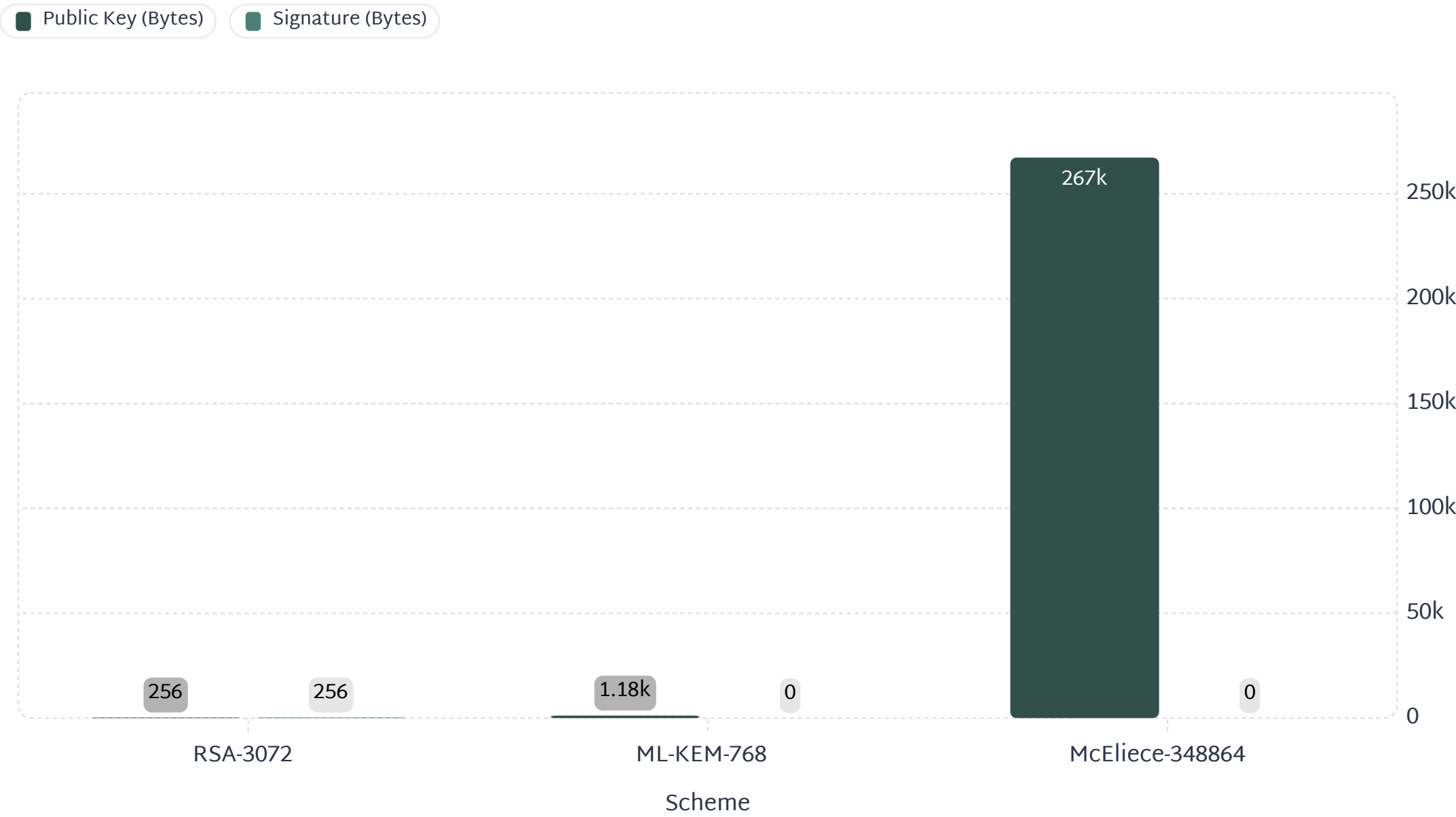
PQC schemes often trade larger keys for stronger quantum resistance.

Key and Signature Sizes

Scheme	Public Key Size	Signature Size
RSA-3072	256 B	256 B
ML-KEM-768	1184 B	—
McEliece-348864	261 KB	—

RSA keys are compact. PQC schemes, especially McEliece, can be much larger. ML-KEM and McEliece are KEMs, so they do not produce signatures.

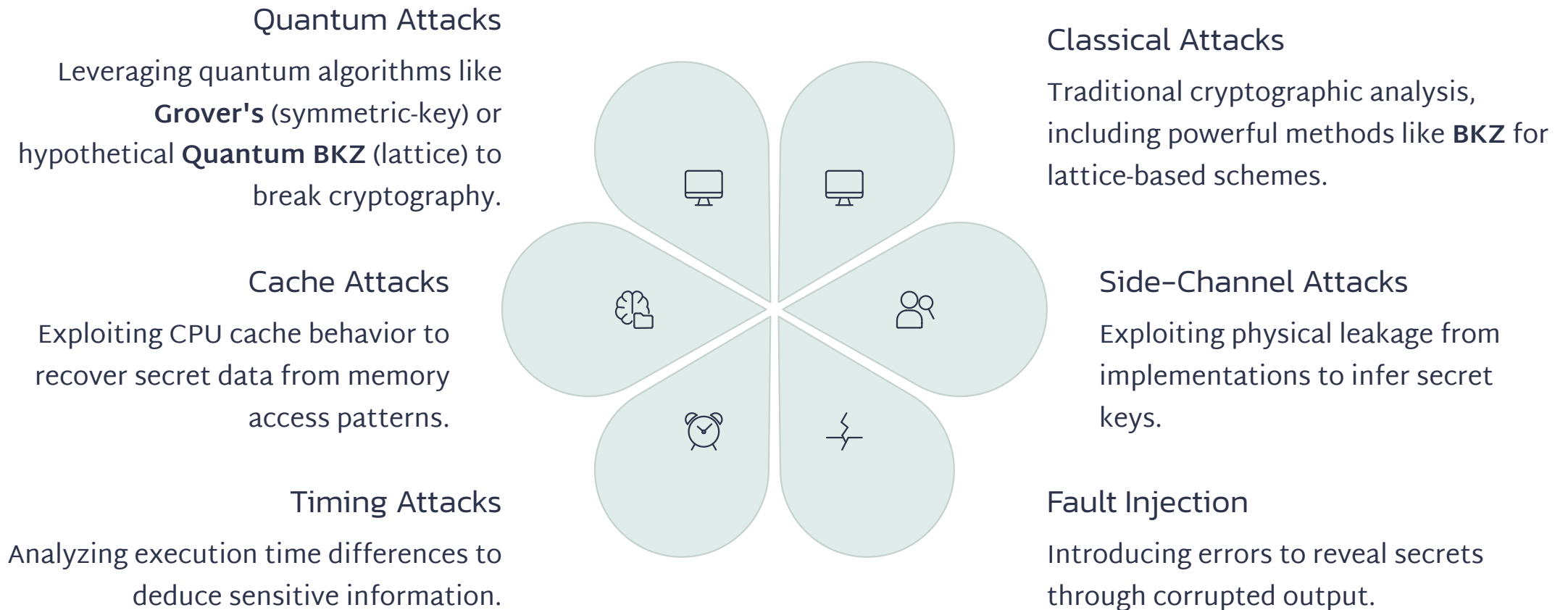
Visualizing Size Differences



The chart highlights the sharp increase in public key size for some PQC schemes.

Attack Models & Implementation Risks

PQC systems must be robust against a spectrum of threats, from theoretical attacks to practical implementation flaws.



❗ The **Constant-Time Requirement** is a critical countermeasure to prevent many side-channel attacks by ensuring cryptographic operations always take a predictable amount of time and resources, regardless of secret data.

"Implementation security is as critical as cryptographic strength in PQC deployment."

Key Challenges in PQC Deployment & Diversity Risk

Although PQC is necessary for quantum-era security, deploying it at scale introduces significant technical and operational challenges.



Large Keys & Signatures
More bandwidth, memory, and storage.



Performance Overheads
Often slower than classical algorithms.



Complex Migration Path
Requires updates across systems and protocols.



New Security Assumptions
Relies on newer, less battle-tested math.

 If lattice assumptions are broken, a large portion of PQC could become insecure.



Code-Based Crypto
Security from error-correcting codes.



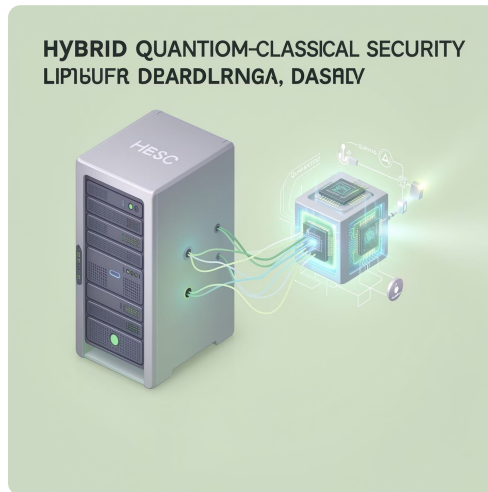
Hash-Based Crypto
Built on well-understood hash functions.



Isogeny-Based Crypto
Uses a distinct approach beyond lattices.

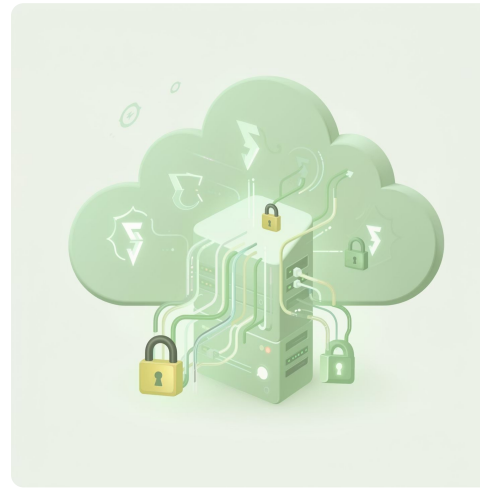
Future Outlook: Trends & Innovation in PQC

Post-quantum cryptography is evolving toward faster, more private, and more scalable quantum-resistant security.



Future Technologies

- Hybrid quantum-classical security
- Privacy-preserving authentication
- Quantum-safe key exchange
- Faster PQC with FPGAs and ASICs



Fully Homomorphic Encryption (FHE)

FHE enables computation on encrypted data without decryption.

Applications:

- Secure cloud computing
- Privacy-preserving medical analysis
- Encrypted AI inference

Key schemes: BFV, CKKS, TFHE.



Future Research Directions

- Stronger theoretical security
- Improved efficiency
- Reduced computation cost
- Proven correctness and security
- Lightweight PQC for IoT

These advancements will enable secure, efficient, and scalable quantum-resistant systems.

CONCLUSION: Embracing Quantum Readiness

The quantum shift demands action now:

01

Quantum Threat

RSA and ECC are vulnerable to quantum attacks.

02

PQC Defense

Post-Quantum Cryptography protects data and communications.

03

Lattice Leadership

Lattice-based schemes lead today's standards.

04

Hard Transition

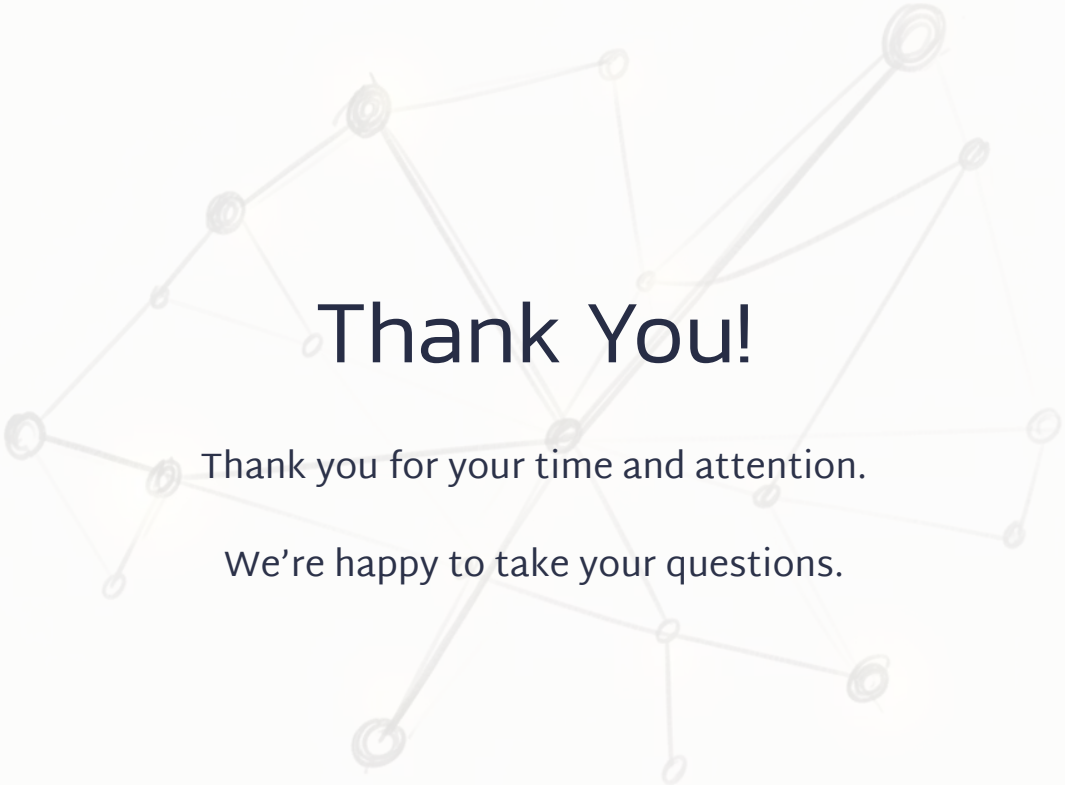
Global PQC deployment is complex but essential.

05

Ongoing Research

PQC and FHE continue to evolve rapidly.

"Cybersecurity depends on post-quantum readiness."



Thank You!

Thank you for your time and attention.

We're happy to take your questions.